- Electric field inside a conductor is zero. The electric field at the surface of the conductor is normal to the surface and has magnitude $E = \frac{\sigma}{\epsilon_0}$.
- The surface of the conductor has the same potential, at all points on the surface.
- Conductor can be charged using the process of induction.
- A dielectric or insulator has no free electrons. When an electric field is applied, the dielectric is polarised.
- Capacitance of a conductor is given by $C = \frac{Q}{V}$.
- Capacitance of a parallel plate capacitor: $C = \frac{\epsilon_0 A}{d}$.
- Electrostatic energy stored in a capacitor: $U = \frac{1}{2}CV^2$.
- The equivalent capacitance for parallel combination is equal to the sum of individual capacitance of the capacitors.
- For a series combination: The inverse of equivalent capacitance is equal to sum of inverse of individual capacitance of capacitors.
- The distribution of charges in the conductors depends on the shape of conductor. For sharper edge, the surface charge density is greater. This principle is used in the lightning arrestor.
- Van de Graaff generator is used to produce large potential difference (~10^7 V).

12th Physics_Vol-1_Unit 1_English Medium.indd 69 08-12-2022 11:04:21